

Numpy 3 Hacker Rank

1.

Task

You are given a 2-D array with dimensions $N \times M$.

Your task is to perform the *sum* tool over axis **0** and then find the *product* of that result.

Input Format

The first line of input contains space separated values of N and M .

The next N lines contains M space separated integers.

Output Format

Compute the sum along axis **0**. Then, print the product of that sum.

Sample Input

```
2 2
1 2
3 4
```

Sample Output

```
24
```

Explanation

The sum along axis **0** = **[4 6]**

The product of this sum = **24**

```

1  # Enter your code here. Read input from STDIN. Print output to STDOUT
2  import numpy as np
3
4  n,m = map(int,input().strip().split())
5  arr=[]
6
7  for _ in range(n):
8      arr.append(list(map(int,input().strip().split())))
9
10 a = np.array(arr)
11
12 print(np.prod(np.sum(a,axis=0)))

```



You have earned 20.00 points!

You are now 65 points away from the gold level for your python badge.



Congratulations

You solved this challenge. Would you like to challenge your friends?



Next Challenge

✓ Test case 0

Compiler Message

Success

✓ Test case 1

✓ Test case 2

Input (stdin)

[Download](#)

```

1  2 2
2  1 2
3  3 4

```

Expected Output

[Download](#)

```

1  24

```

2.

Task

You are given a 2-D array with dimensions $N \times M$.

Your task is to perform the min function over axis 1 and then find the max of that.

Input Format

The first line of input contains the space separated values of N and M .

The next N lines contains M space separated integers.

Output Format

Compute the min along axis 1 and then print the max of that result.

Sample Input

```
4 2
2 5
3 7
1 3
4 0
```

Sample Output

```
3
```

Explanation

The min along axis 1 = $[2, 3, 1, 0]$

The max of $[2, 3, 1, 0] = 3$

```
1 # Enter your code here. Read input from STDIN. Print output to STDOUT
2 import numpy as np
3 n,m = map(int,input().strip().split())
4 arr=[]
5 for _ in range(n):
6     arr.append(list(map(int,input().strip().split(' '))))
7
8 a = np.array(arr)
9 print(np.max(np.min(a,axis=1)))
```



You have earned 20.00 points!

You are now 45 points away from the gold level for your python badge.



Congratulations

You solved this challenge. Would you like to challenge your friends?

[Next Challenge](#)

✓ Test case 0

Compiler Message

✓ Test case 1

Success

✓ Test case 2

Input (stdin)

[Download](#)

```
1 4 2
2 2 5
3 3 7
4 1 3
5 4 0
```

Expected Output

[Download](#)

```
1 3
```

3.

Task

You are given a 2-D array of size $N \times M$.

Your task is to find:

1. The mean along axis **1**
2. The var along axis **0**
3. The std along axis *None*

Input Format

The first line contains the space separated values of N and M .

The next N lines contains M space separated integers.

Output Format

First, print the mean.

Second, print the var.

Third, print the std.

Sample Input

```
2 2
1 2
3 4
```

Sample Output

```
[ 1.5  3.5]
[ 1.  1.]
1.11803398875
```

```

1  # Enter your code here. Read input from STDIN. Print output to STDOUT
2  import numpy as np
3
4  n,m =map(int,input().split())
5
6  arr=[]
7  for _ in range(n):
8      arr.append(list(map(int,input().split())))
9
10 a = np.array(arr)
11 print(np.mean(a, axis=1))
12 print(np.var(a, axis=0))
13 print(round(np.std(a),11))
14

```



You have earned 20.00 points!

You are now 25 points away from the gold level for your python badge.

86%

375/400

Congratulations

You solved this challenge. Would you like to challenge your friends?



Next Challenge

Test case 0

Compiler Message

Success

Test case 1

Test case 2

Input (stdin)

Download

```

1  2 2
2  1 2
3  3 4

```

Expected Output

Download

```

1  [1.5 3.5]
2  [1. 1.]
3  1.11803398875

```